

EKSAMENSOPPGÅVE/EKSAMENSOPPGAVE

Emnekode: MAS 149

Emnenamn/Emnenavn: STATIKK OG FASTHETSLÆRE

Utdanning/kull/klasse: Diverse

Dato: 30.09.2024

Eksamensform: Skriftlig

Eksamenstid: 2 timer

Tal på eksamensoppgåver/Antall eksamensoppgaver: 4

Tal på sider (medrekna denne)/Antall sider (inkludert denne): 10

Tal på vedlegg/Antall vedlegg: 2

Tillatne hjelpemiddel/Tillatte hjelpemidler: Enkel kalkulator

(ikke programmerbar)

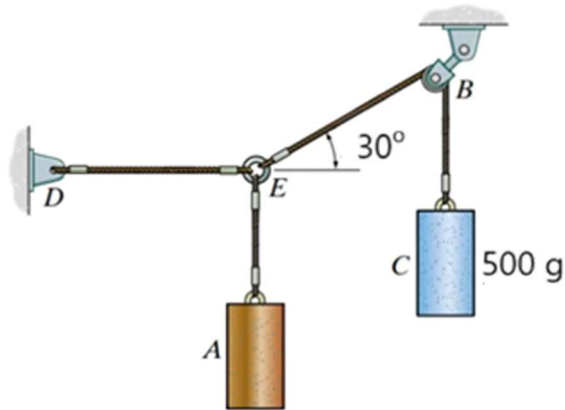
Fagansvarleg/Fagansvarlig: Javier de Elio Medina (Haugesund)

English version

Exercise 1 (20%):

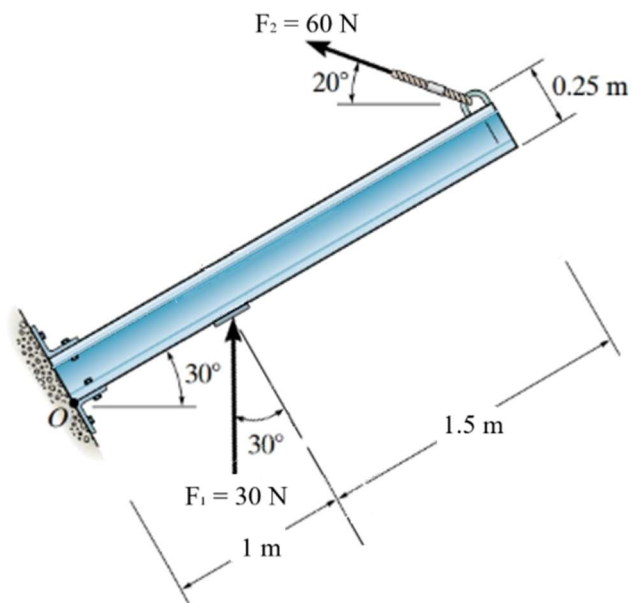
If the mass of cylinder C is 500 g, determine:

- The mass of cylinder A in order to hold the assembly in the position shown in the figure.
- The tension in the cable DE.



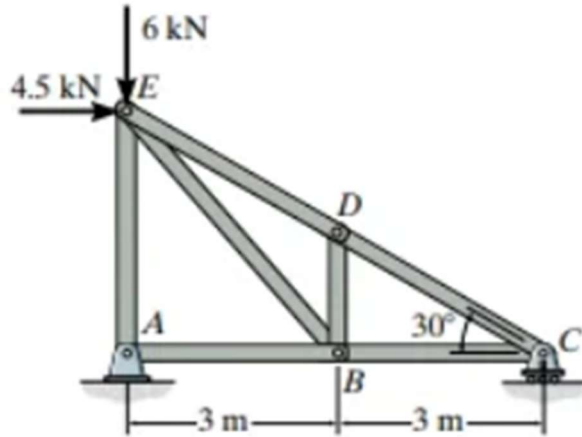
Exercise 2 (20%)

- Determine the moment of the force F_1 acting on the beam shown in the figure about point O.
- Determine the moment of the force F_2 acting on the beam shown in the figure about point O.



Exercise 3 (30%)

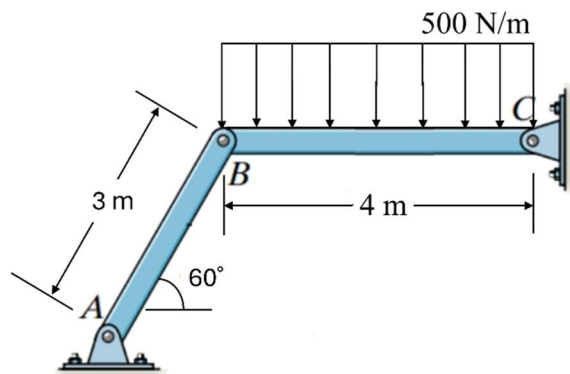
The truss in the figure has a fixed support (pin support) at point A and a roller support at point C. Determine the forces in all members and explain which of them are subjected to tensile or compressive forces.



Exercise 4 (30%)

For the frame shown in the figure:

- Draw the Free-Body diagram (FBD) of AB and BC.
- Determine the reactions at pin A, B, and C.

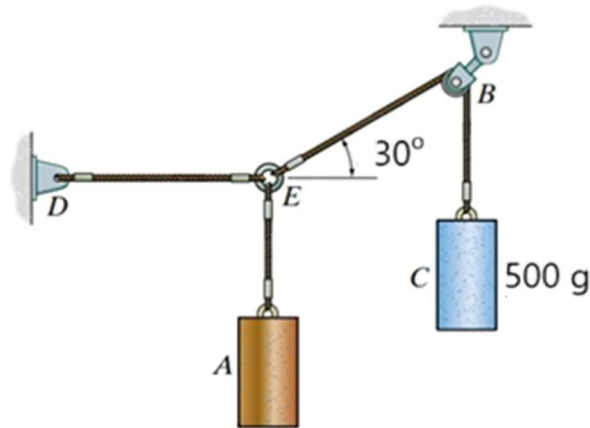


Bokmål versjon

Oppgave 1 (20%):

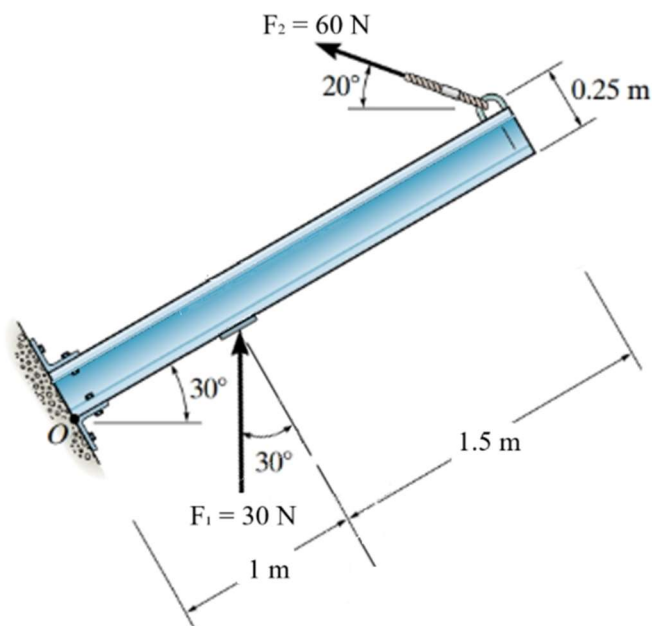
Hvis massen til sylinder C er 500 g, bestem:

- Massen til sylinder A for å holde arrangementet i posisjonen vist i figuren.
- Kraften i kabelen DE.



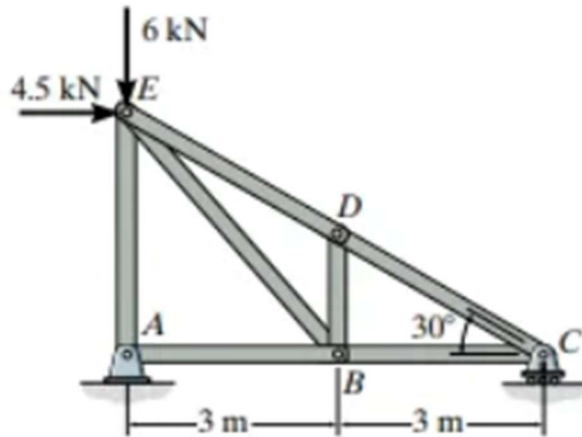
Oppgave 2 (20%)

- Bestem kraftmomentet til kraften F_1 som virker på bjelken vist i figuren om punkt O.
- Bestem kraftmomentet til kraften F_2 som virker på bjelken vist i figuren om punkt O



Oppgave 3 (30%)

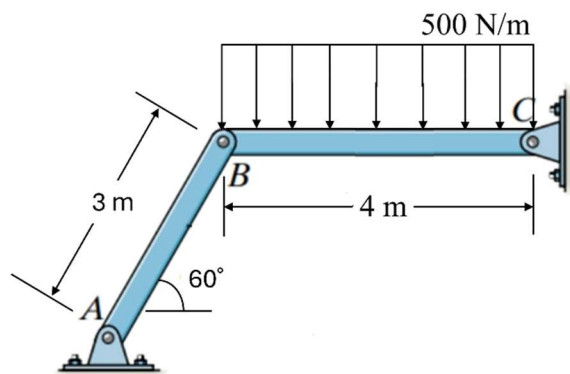
Fagverket på figuren har et fastlager (boltlager) i punkt A og et rullelager i punkt C. Bestem kreftene i alle staver og forklar hvilke av dem som er utsatt for strekk eller trykkrefter.



Oppgave 4 (30%)

For rammeverket vist i figuren:

- c) Tegn fritt legeme diagrammet for AB og BC
- d) Bestem reaksjonskreftene i punkt A, B og C (boltlager).

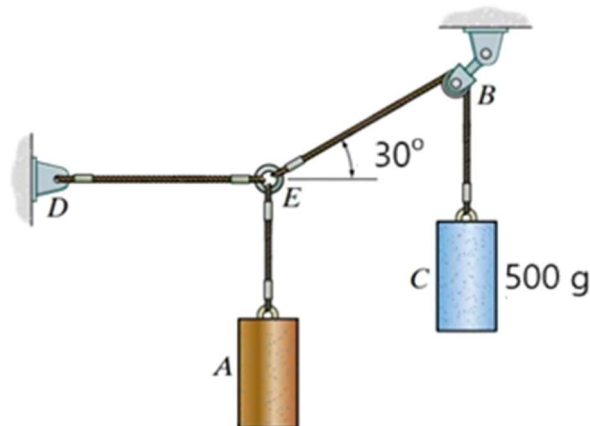


Nynorsk versjon

Oppgåve 1 (20%):

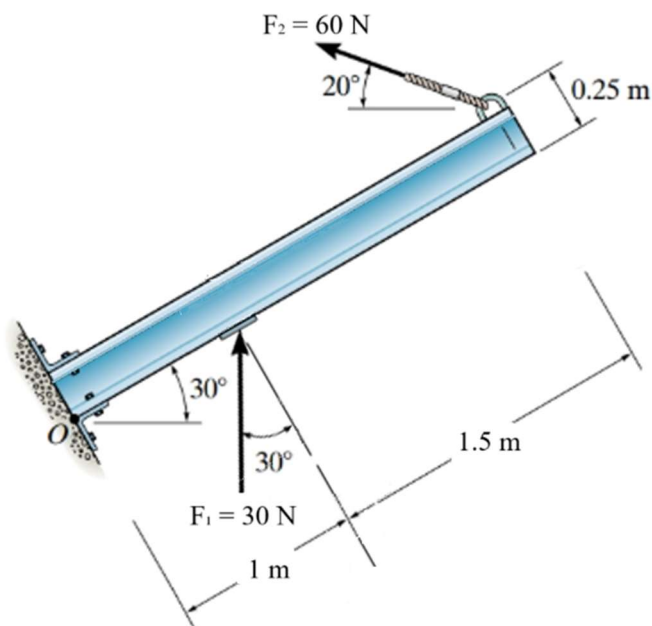
Dersom massen til sylinder C er 500 g, finn:

- Massen sylinder A må ha for å halda sylindrar og kablar i posisjonen som vist i figuren.
- Krafta i kabelen DE.



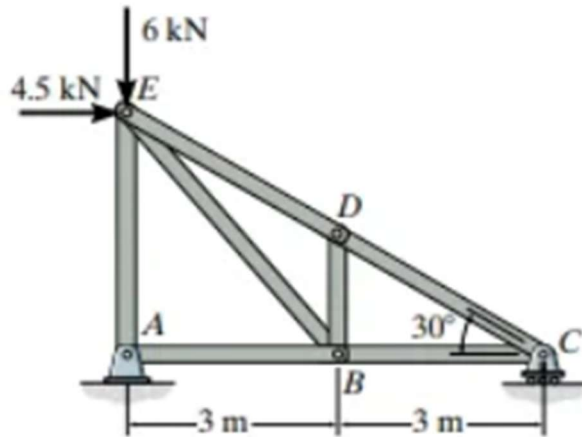
Oppgåve 2 (20%):

- Finn kraftmomentet til krafta F_1 som verkar på bjelken vist i figuren om punkt O.
- Finn kraftmomentet til krafta F_2 som verkar på bjelken vist i figuren om punkt O.



Oppgave 3 (30%)

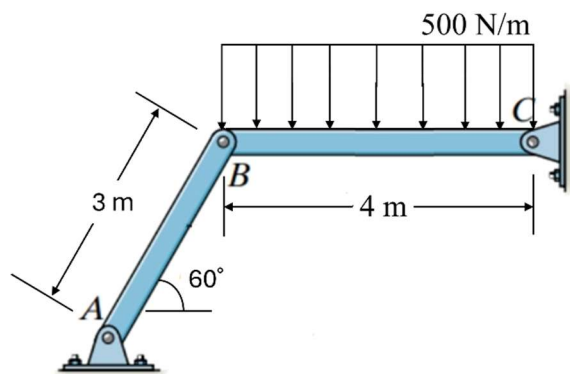
Fagverket på figuren har eit fastlager (boltlager) i punkt A og eit rullelager i punkt C. Bestem kreftene i alle stavar og forklar kva for nokre av dei som er utsette for strekk eller trykkrefter.



Oppgave 4 (30%)

For rammeverket vist i figuren:

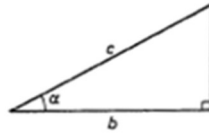
- Teikn frilekamdiagrammet for AB og BC
- Finn reaksjonskreftene i punkt A, B og C (boltlager).



TRIGONOMETRI

Pytagoreiske
læresetning

$$c^2 = a^2 + b^2$$



c hypotenus
 a |
 b kateter

Trigonometriske
funksjoner

$$\sin \alpha = \frac{a}{c}$$

$$\cos \alpha = \frac{b}{c}$$

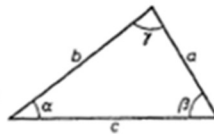
$$\tan \alpha = \frac{a}{b}$$

Sinussetningen

$$\frac{a}{\sin \alpha} = \frac{b}{\sin \beta} = \frac{c}{\sin \gamma}$$

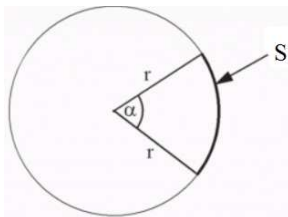
Cosinussetningen

$$a^2 = b^2 + c^2 - 2bc \cos \alpha$$



Tangenssetningen

$$\frac{a-b}{a+b} = \frac{\tan \frac{\alpha-\beta}{2}}{\tan \frac{\alpha+\beta}{2}}$$



$$S = \alpha r \quad \alpha [\text{rad}]$$

Geometric Properties of Area Elements

Material Property Relations

Poisson's ratio

$$\nu = -\frac{\epsilon_{\text{lat}}}{\epsilon_{\text{long}}}$$

Generalized Hooke's Law

$$\epsilon_x = \frac{1}{E} [\sigma_x - \nu(\sigma_y + \sigma_z)]$$

$$\epsilon_y = \frac{1}{E} [\sigma_y - \nu(\sigma_x + \sigma_z)]$$

$$\epsilon_z = \frac{1}{E} [\sigma_z - \nu(\sigma_x + \sigma_y)]$$

$$\gamma_{xy} = \frac{1}{G} \tau_{xy}, \gamma_{yz} = \frac{1}{G} \tau_{yz}, \gamma_{zx} = \frac{1}{G} \tau_{zx}$$

where

$$G = \frac{E}{2(1 + \nu)}$$

Relations Between w , V , M

$$\frac{dV}{dx} = w(x), \quad \frac{dM}{dx} = V$$

Elastic Curve

$$\frac{1}{\rho} = \frac{M}{EI}$$

$$EI \frac{d^4 v}{dx^4} = w(x)$$

$$EI \frac{d^3 v}{dx^3} = V(x)$$

$$EI \frac{d^2 v}{dx^2} = M(x)$$

Buckling

Critical axial load

$$P_{cr} = \frac{\pi^2 EI}{(KL)^2}$$

Critical stress

$$\sigma_{cr} = \frac{\pi^2 E}{(KL/r)^2}, \quad r = \sqrt{I/A}$$

Secant formula

$$\sigma_{max} = \frac{P}{A} \left[1 + \frac{ec}{r^2} \sec \left(\frac{L}{2r} \sqrt{\frac{P}{EA}} \right) \right]$$

Energy Methods

Conservation of energy

$$U_e = U_i$$

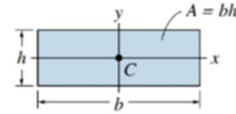
Strain energy

$$U_i = \frac{N^2 L}{2AE} \quad \text{constant axial load}$$

$$U_i = \int_0^L \frac{M^2 dx}{2EI} \quad \text{bending moment}$$

$$U_i = \int_0^L \frac{f_s^2 V^2 dx}{2GA} \quad \text{transverse shear}$$

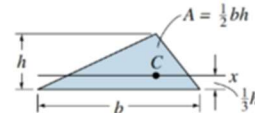
$$U_i = \int_0^L \frac{T^2 dx}{2GJ} \quad \text{torsional moment}$$



Rectangular area

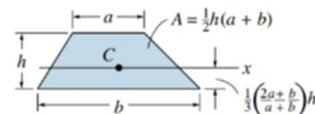
$$I_x = \frac{1}{12} b h^3$$

$$I_y = \frac{1}{12} h b^3$$

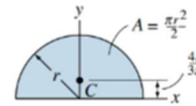


Triangular area

$$I_x = \frac{1}{36} b h^3$$



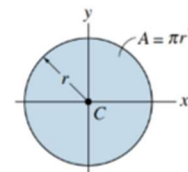
Trapezoidal area



Semicircular area

$$I_x = \frac{1}{8} \pi r^4$$

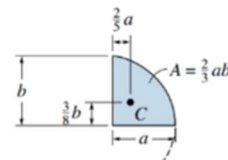
$$I_y = \frac{1}{8} \pi r^4$$



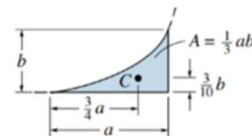
Circular area

$$I_x = \frac{1}{4} \pi r^4$$

$$I_y = \frac{1}{4} \pi r^4$$



Semiparabolic area



Exparabolic area